

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 9/16/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Yes, team was able to meet over teams with all members attending
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. Yes we have met with the team sponsor.
 - i. Ian in Lab, David and John in communication over Teams at 5:30 PM 9/17
3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. We were able to connect the Unitree to amobile hotspot. However, this is not strong enough to download any repositories needed for our Yolo model.
 - b. The camera source has been found in the directory and we have figured out how to shut down this process. We can record video files and save them. The next step is generating a live feed of the camera process
4. Describe the tasks to be completed the coming week.
 - a. Still testing the command ability of the robot over code for decision making as well
 - b. In the coming week, hopefully we will have a live feed available for the robot. When this live feed is available, we will be able to put our tracking model on top of it.
 - c. Getting a stable Internet connection will also allow for our model to run effectively. One of the ways planned is hopefully to set up a stronger access point the RaspberryPi to connect to

Copy the following table from the last week's report (09/02)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons	2	0	2
John Sinfuego			

Fill in the following table for this week: (09/09)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	0	2	4
Ian Simmons	2	0	2
John Sinfuego	2	0	2

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: Ian Simmons

Report Date: 9/16/2024

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **before the past week**:

1. **Week 2, Task 1 (9/9)**
 - a. Research ways of adding Wi-Fi to robot
 - i. Access the Pi's config directly
 - ii. We were able to access the config settings of the Raspberry Pi and add the university network. However this did not give us internet access
2. **Week 2, Task 2 (9/9)**
 - a. Connect to RaspberryPi and explore directories
 - i. The goal of this is to familiarize ourselves with the procedure for navigating through the libraries stored on the robot whilst practicing Linux commands at the same time.
 - ii. Located a directory called UnitreeCameraSDK and this appears to have a way of generating live feed from the camera in some way. Will be continuing down this direction

For **each** task you have completed **during the past week**:

3. **Week 3, Task 1 (9/16)**
 - a. Explore files within robot for usability in network access
 - i. When adjusting the configuration settings in the Raspberry Pi of the dog, although connecting to the university network was not possible, we were able to connect to a mobile hotspot.
 - ii. The robot connected to the Internet, but we need to find a way to set up a stronger access point for testing capabilities.
4. **Week 3, Task 2 (9/16)**
 - a. Find/Register Mac address to local network
 - i. Unfortunately, Registering the Mac address was only possible to FAU guest Wi-Fi. However, it still does not work as a certificate still needs to be obtained to access the University internet.

- ii. Still looking into the wireless access point at this moment.

For **each** task you are currently working on or plan to work on in the next period:

1. **Week 4, Task 1 (9/23)**
 - a. This next week my focus is on completing the process of accessing the cameras through the CameraSDK. And building the environment necessary on the Raspberry Pi.
2. **Week 4, Task 2 (9/23)**
 - a. Currently I am downloading the files and directories needed to stream the camera process to the local computer
 - b. The issue is that these directories are all incorporated and are likely to run into small errors during execution

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: David Uzan

Report Date: 9/16/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the before the past week**:

1. Task number/name

General Task 3 (9-13-24) – Establish baseline for use cases and testing and search for methods to run YOLO directly on Uni-Go1

Personal Week 3 (9-13-24) – Migrate object detection from Google Colab to local IDE and test live feedback capabilities of the Uni-Go1

For **each** task you have completed **during the past week**:

1. Task number/name

N/A

2. Short description – these descriptions form the basis of the final project report
 - a. Implementation, testing, results

N/A

- b. Brief and to point

N/A

3. Outcome of the task activity

- a. How did the task advance your project?

N/A

- b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

N/A

For **each** task you are currently working on or plan to work on in the next period:

- 1. Task number/name

General Task 4 (9-23-24) – Incorporate the model into Uni-Go1 and run tests across multiple cameras.

Personal Week 4 (9-23-24) – Locate camera source directory in Uni-Go1 and display live video feed from it.

Personal Week 5 (9-23-24) - Add object detection python file to Uni-Go1.

- 2. Short description

- a. Plans, resources needed, implementation, results

Using the manual of the Uni-Go1 and previous projects' work to access storage locations inside of the robot are crucial for the next step of implementation. I have found possible save locations but require further testing to discern if it is the proper site for camera operations. My group mates have also begun installation of necessary software, and I will run basic tests to get a more definitive answer of how the robot will detect objects in its environment.

b. Main challenges, reasons for any delay in completion

A previous project using the same Uni-Go1 model found that accessing the cameras of the robot killed the process until it was reset. The directories mentioned previously could also be temporary folders that have no workflow. We would have to spend more time figuring out a way that does not break the camera feed from the robot to increase reliability of the system.

3. Outcome of the task activity

- a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

```
camera_config = {
    'front': {'nano_ip': '192.168.123.13', 'device': '/dev/video1'},
    'chin': {'nano_ip': '192.168.123.13', 'device': '/dev/video0'},
    'right': {'nano_ip': '192.168.123.14', 'device': '/dev/video0'},
    'left': {'nano_ip': '192.168.123.14', 'device': '/dev/video1'},
    'bottom': {'nano_ip': '192.168.123.15', 'device': '/dev/video0'}
}
```

```
save_directory):
if quality == 'y4m':
    ffmpeg_command = f"ffmpeg -f video4linux2 -i {device} -t {duration} -r
    {frame_rate} -pix_fmt yuv420p -f yuv4mpegpipe
    /home/unitree/{file_name}.y4m -vstats_file
    /home/unitree/{file_name}.log"
elif quality == 'mp4':
    ffmpeg_command = f"ffmpeg -f video4linux2 -i {device} -t {duration} -r
    {frame_rate} -pix_fmt yuv420p -c:v libx264 -preset ultrafast
    /home/unitree/{file_name}.mp4 -vstats_file
    /home/unitree/{file_name}.log"
else:
    raise ValueError("Invalid video quality specified.")
```

File paths accessed in the code to capture video and record file statistics.

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: John Angelo Sinfuego

Report Date: 9/16/24

For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 3 **9-9-24**- Establish Baseline for use cases and testing

Personal Week 3 **9-9-24** - Search for methods to run YOLOv8 directly from go1

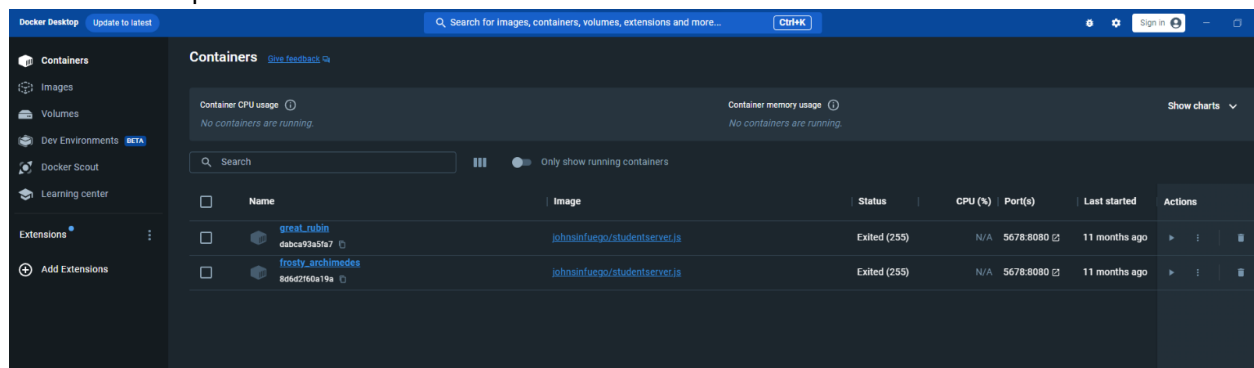
2. Short description

We have to run the YOLOv8 model directly from the unigo1 robot in order to establish object detection with our live feed. Since this, we have been researching different methods in order to figure out how to accomplish this task.

b. There is no wifi on the unigo1 robot, which makes it challenging to download the YOLOv8 repository into it because this needs wifi in order to be done. I have researched more methods to accomplish this, via different wifi adapters to put in with raspberry pi, and using a program called docker in order to establish code into the raspberry pi so it can be transferred into go1.

3. Outcome of the task activity

Docker desktop-



Name of the Member: John Angelo Sinfuego

Report Date: 9/16/24

For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 5 9-16-24- Incorporate our model onto unitree, Run tests in order to access camera

Personal Week 5 9-16-24 - Search for methods to run YOLOv8 directly from go1, Look into UniTree ROS(Video/Movement), Gather Software/code resources for potential use.

2. Short description

a. We must incorporate our YOLOv8 model into unitree so we can access the yolo repository into our robot's live footage as well as run tests so we can access our camera. This is found within the files in unigo1. We have found a repository which could be useful for retrieving video and movement from unigo1. The UniTree ROS for the unigo1 robot which utilizes video/movement. We are going to be using UniTree ROS1 since ROS2 is not compatible with unigo1. This repository utilizes simulation packages for unitree robots and could be very useful in the future.

b. For our challenges, we still have trouble incorporating our YOLOv8 repository into unigo1. Since our dog does not have wi-fi, we still have to find a solution for this and are still researching it. We figured out that

3. Outcome of the task activity

Repository for unitree ROS1

unitree_rosPublic

Watch21

master2 Branches0 Tags

Go to file

Add file

Code

Unknownadd go2 rotor inertial to g2's urdfa615577 · last week108 Commits

robots	add go2 rotor inertial to g2's urdf	last week
unitree_controller	fixed IMU, removed rpy from IMU	4 years ago
unitree_gazebo	Update z1.launch	2 years ago
unitree_legged_control	add the source file of libunitree_joint_control_tool	2 months ago
unitree_ros_to_real @ b989870	Updare unitree_ros_to_real	last year
.gitignore	modified readme for v3.2.1	3 years ago
.gitmodules	add unitree_ros_to_real submodule	last year
LICENSE	add Z1	2 years ago
README.md	modify z1_urdf	last year

READMEBSD-3-Clause license

Introduction

Here are the ROS simulation packages for Unitree robots, You can load robots and joint controllers in Gazebo, so you can perform low-level control (control the torque, position and angular velocity) of the robot joints. Please be aware that the Gazebo simulation cannot do high-level control, namely walking. Aside from these simulation functions, you can also control your real robots in ROS with the [unitree_ros_to_real](#) packages. For real robots, you can do high-level and low-level control using our ROS packages.

Packages:

Robot description: `go1_description` , `a1_description` , `aliengo_description` , `laikago_description` , `z1_description`

Robot and joints controller: `unitree_controller` , `z1_controller`

For **each** task you are currently working on or plan to work on in the next period:

2. Task number/name

General Task 5 9-23-24- Run tests to access camera

Personal Week 5 9-23-24- - Kill nano1 camera process on dog in python, create a function to start new live feed on camera, Gather software/code resources for potential use

2. Short description

a. We have to kill the nano1 camera process on the dog and we will do this using python. We have to kill this process because it is required to run the YOLOv8 repository on unigo1 so we can retrieve object detection through the dog. I will also be looking into creating a function to start new live feed within the camera. I have looked into the untree_ros repository and there are some code that can be useful. Specifically code that we can use to control the robot using gazebo.

b. We are still running tests to access the camera, still looking for potentially more code that can help with this as well as still finding a way to incorporate wi-fi within the dog so we can upload YOLOv8 through it

4. Outcome of the task activity

Here is some code we can potentially use within the repository to control the robot using Gazebo

unitree_gazebo & unitree_controller:

You can launch the Gazebo simulation with the following command:

```
roslaunch unitree_gazebo normal.launch rname:=a1 wname:=stairs
```

Where the `rname` means robot name, which can be `laikago`, `aliengo`, `a1` or `go1`. The `wname` means world name, which can be `earth`, `space` or `stairs`. And the default value of `rname` is `laikago`, while the default value of `wname` is `earth`. In Gazebo, the robot should be lying on the ground with joints not activated.

1. Stand controller

After launching the gazebo simulation, you can start to control the robot:

```
roslaunch unitree_controller unitree_servo
```

And you can add external disturbances, like a push or a kick:

```
roslaunch unitree_controller unitree_external_force
```

2. Position and pose publisher

Here we demonstrated how to control the position and pose of robot without a controller, which should be useful in SLAM or visual development.

Then run the position and pose publisher in another terminal:

```
roslaunch unitree_controller unitree_move_kinetic
```

The robot will turn around the origin, which is the movement under the world coordinate frame. And inside of the source file [move_publisher.cpp](#), we also provide the method to move using the robot coordinate frame. You can change the value of `def_frame` to `coord::ROBOT` and run the catkin_make again, then the `unitree_move_publisher` will move robot under its own coordinate frame.

z1_controller